

Revised Summer Plan (June 16 - July 20, 2026) | 34 Days

Week 1 is gone. No guilt. This plan starts NOW.

First: The Anxiety Answer

“Java Fullstack vs Backend + Cloud - which should I invest in?”

They are NOT competing paths. They are the SAME path with 80% overlap.

Java Fullstack (Anudip) = Java + Spring Boot + Databases + Frontend basics
Backend + Cloud (Nimblize) = Java + Spring Boot + Databases + Cloud deployment

OVERLAP = Java + Spring Boot + Databases (80%)

Skill	Fullstack?	Backend+Cloud?	Placements?	AKTU?
Java (core language)	YES	YES	YES	YES
Spring Boot (backend framework)	YES	YES	YES	-
Databases / SQL	YES	YES	YES	YES (DBMS)
REST APIs	YES	YES	YES	-
Frontend (HTML/CSS/JS)	YES	minimal	helpful	-
Cloud (AWS/Azure)	-	YES	bonus	-

VERDICT: Learn Java + Spring Boot + Databases. That IS both fullstack AND backend. Cloud and frontend are add-ons you layer on top. Stop thinking of them as separate paths.

Will you have a full-fledged skill by Jan 2027? YES - if you focus on the overlap (Java + Backend + DB) instead of panicking about everything. You have 7 months. Most people learn placement-ready skills in 3-4 months of focused effort.

Your Current Commitments (Sorted by Deadline)

#	What	Deadline	Effort	Status
1	Nimblize Phase 2 Report (Backend + Cloud)	Sun June 22	~8-10 hrs	NOT STARTED
2	Trilogy DSA Mock Test (filtering round)	Tue June 23	~15-20 hrs prep	NOT STARTED
3	College Daily DSA Challenge	Daily	30-60 min/day	ONGOING
4	TransitPulse Database Design	July 10	~15-20 hrs	NOT STARTED
5	Java mastery (GFG)	Ongoing	woven into daily	IN PROGRESS
6	LeetCode/HackerRank profiles	Ongoing	part of DSA practice	STARTED
7	Spring Boot + Project	Before semester	after July 10	NOT STARTED
8	DAA prep (C++)	Before semester	last week	NOT STARTED

The Key Insight: Everything Feeds Everything

Nimblize Report (APIs + DB design)

- > directly teaches you DBMS concepts

- > helps TransitPulse database

- > helps AKTU 5th sem DBMS subject

- > helps placement interviews

Trilogy DSA prep

- > same problems as college daily challenge
- > builds LeetCode/HackerRank profile
- > directly tests placement readiness

TransitPulse DB design

- > teaches real-world database skills
- > is your portfolio project
- > proves skills to Nimblize

Nothing here is wasted effort. Every task feeds at least 2 other goals.

PHASE 1: URGENT FIRES (June 16-23 | 8 days)

Priority: Nimblize Report + Trilogy DSA Test

Nimblize Report Plan (Due Sunday June 22, 7 PM)

You need to cover Backend + Cloud. Here's how to structure your report:

Backend Section:

1. APIs - What are REST APIs, HTTP methods (GET/POST/PUT/DELETE), JSON, how Nimblize frontend will talk to backend
2. Database Design - Entities for Nimblize (Users, Products, Orders, Analytics), ER diagrams, SQL vs NoSQL choice
3. Architecture Mapping - Draw a diagram showing: Frontend <-> REST API <-> Backend (Spring Boot) <-> Database, plus how Analytics/Blockchain/Cybersecurity modules connect

Cloud Section:

1. Cloud Fundamentals - AWS vs Azure vs GCP comparison, what services Nimblize would use (EC2, RDS, S3, Lambda)
2. Infrastructure Design - How to handle traffic growth (load balancers, auto-scaling, CDN)
3. Deployment Planning - Infrastructure diagram for Nimblize (draw.io or Excalidraw)

Time allocation: ~2 hrs/day for 4 days = 8 hrs total

Trilogy DSA Mock Test Topics (Tue June 23)

Since I cannot access your other chat's topic list, here are the standard DSA topics that CodeNation/Trilogy typically tests at this level:

Topic	Key Concepts	Must-Solve LeetCode
Arrays	Traversal, two pointers, prefix sum	LC #1 Two Sum, LC #53 Max Subarray
Strings	Manipulation, palindromes, anagrams	LC #125 Valid Palindrome, LC #242 Valid Anagram
Linked Lists	Traversal, cycle detection, reversal	LC #206 Reverse Linked List, LC #141 Linked List Cycle
Stacks/Queues	Implementation, balanced brackets	LC #20 Valid Parentheses, LC #155 Min Stack
Sorting	Bubble, Selection, Merge, Quick sort	LC #912 Sort an Array
Searching	Linear search, Binary search	LC #704 Binary Search, LC #35 Search Insert Position
Recursion	Base case, factorial, fibonacci	LC #509 Fibonacci Number, LC #70 Climbing Stairs
Basic Math	GCD, prime numbers, modular arithmetic	LC #1979 Find GCD of Array
Hashing	HashMap, frequency counting	LC #217 Contains Duplicate, LC #1 Two Sum
Time Complexity	Big-O notation for all above	Theory - no LC needed

IMPORTANT: You said you didn't understand Linked List Cycle. Here's the concept in 30 seconds:

Linked List Cycle uses Floyd's Tortoise and Hare algorithm. Two pointers: slow moves 1 step, fast moves 2 steps. If they ever meet, there's a cycle. If fast reaches null, no cycle. That's it.

Week Schedule: June 16-22

Day	Morning (2 hrs)	Afternoon (2 hrs)	Evening (1 hr)
Mon 16 (Today)	Nimblize: Research APIs + take notes	DSA: Arrays + Hashing theory	College DSA challenge + LC #1 Two Sum
Tue 17	Nimblize: Database Design section + ER diagram	DSA: Linked Lists (understand cycle, reversal)	College DSA + LC #206 Reverse Linked List
Wed 18	Nimblize: Architecture diagram (use draw.io)	DSA: Stacks/Queues + Sorting theory	College DSA + LC #20 Valid Parentheses
Thu 19	Nimblize: Cloud Fundamentals section	DSA: Searching + Binary Search	College DSA + LC #704 Binary Search
Fri 20	Nimblize: Infrastructure Design + Deployment diagram	DSA: Recursion + Strings	College DSA + LC #125 Valid Palindrome
Sat 21	Nimblize: Compile and polish full report	DSA: Full revision - solve 5 mixed problems	College DSA + LC #53 Max Subarray
Sun 22	Nimblize: Submit report by 7 PM	DSA: Mock test practice - timed problem solving	LC #141 Linked List Cycle (understand it properly)

***ABOUT COLLEGE DSA DAILY CHALLENGE:** Do it in Java. If you can solve it yourself - push to GitHub. If you need AI help to understand - that's fine, but UNDERSTAND it first, then rewrite it yourself, THEN push. Never push code you don't understand.*

PHASE 2: DBMS + DSA GRIND (June 23 - July 6 | 14 days)

Priority: TransitPulse Database + Consistent DSA + Java Deepening

DBMS Learning Path (for TransitPulse + AKTU 5th sem)

You said you know zero about DBMS. Here's the learning order:

Week	DBMS Topics	Resource	TransitPulse Application
Week 1 (June 23-29)	What is DBMS, Tables, Rows, Columns, Primary/Foreign Keys, ER Diagrams, Normalization (1NF, 2NF, 3NF)	GFG DBMS Tutorial (free)	Draw ER diagram for TransitPulse entities
Week 2 (June 30-July 6)	SQL basics (CREATE, INSERT, SELECT, UPDATE, DELETE), JOINS, GROUP BY, Aggregate functions	GFG SQL Tutorial + W3Schools SQL	Write actual SQL for TransitPulse tables

Week 3 Schedule: June 23-29

Day	Morning (2 hrs)	Afternoon (1.5 hrs)	Evening (1 hr)
Mon 23	Post-test review: identify weak topics	DBMS: What is DBMS, Tables, Types of Keys	College DSA + LC #217 Contains Duplicate
Tue 24	Trilogy session	DBMS: ER Diagrams - Entities, Relationships, Cardinality	College DSA + LC #242 Valid Anagram
Wed 25	Java: OOP deep dive (GFG) - Classes, Inheritance	DBMS: Draw ER diagram for TransitPulse	College DSA + LC #121 Best Time to Buy/Sell Stock
Thu 26	Java: Collections (ArrayList, HashMap, HashSet)	DBMS: Normalization (1NF, 2NF, 3NF) with examples	College DSA + LC #88 Merge Sorted Array
Fri 27	Java: Exception Handling, Strings	DBMS: Normalize TransitPulse tables	College DSA + LC #26 Remove Duplicates
Sat 28	Java: Practice - solve 3 Java problems on GFG	Nimblize tasks (if any)	College DSA + LC #35 Search Insert Position
Sun 29	REST DAY	Light review	-

Week 4 Schedule: June 30 - July 6

Day	Morning (2 hrs)	Afternoon (1.5 hrs)	Evening (1 hr)
Mon 30	SQL: CREATE TABLE, INSERT, SELECT basics	Apply SQL to TransitPulse schema	College DSA + LC #509 Fibonacci Number
Tue 1	Trilogy session	SQL: WHERE, ORDER BY, LIMIT, UPDATE, DELETE	College DSA + LC #70 Climbing Stairs
Wed 2	SQL: JOINS (INNER, LEFT, RIGHT)	Write JOIN queries for TransitPulse	College DSA + LC #155 Min Stack
Thu 3	SQL: GROUP BY, HAVING, Aggregate functions	TransitPulse: Complex queries (analytics)	College DSA + LC #75 Sort Colors
Fri 4	Java: Multithreading basics + File I/O	TransitPulse: Finalize DB schema document	College DSA + LC #215 Kth Largest Element
Sat 5	TransitPulse: Create complete DB with all tables in PostgreSQL/MySQL	Test with sample data	College DSA + LC #198 House Robber
Sun 6	REST DAY	Review week	-

PHASE 3: BUILD + DAA PREP (July 7-20 | 14 days)

Priority: TransitPulse DB completion + Spring Boot intro + DAA Unit I

Week 5 Schedule: July 7-13

Day	Morning (2 hrs)	Afternoon (1.5 hrs)	Evening (1 hr)
Mon 7	TransitPulse: Final DB testing + documentation	Spring Boot: Setup + first REST API (Telusko YouTube)	College DSA + LC #55 Jump Game

Tue 8	Trilogy session	Spring Boot: REST Controllers, GET/POST	College DSA + LC #3 Longest Substring Without Repeating
Wed 9	Spring Boot: Connect to PostgreSQL (JPA)	DAA: Asymptotic Notations - Big-O (Abdul Bari)	College DSA + LC #33 Search Rotated Array
Thu 10	TransitPulse DB: SUBMIT / Final delivery	Spring Boot: CRUD operations	College DSA + LC #49 Group Anagrams
Fri 11	Spring Boot: Build a mini REST API project	DAA: Sorting algorithms theory (Abdul Bari)	College DSA + LC #78 Subsets
Sat 12	DAA: Merge Sort + Quick Sort (theory + C++ implementation)	Spring Boot: Practice building APIs	College DSA + LC #46 Permutations
Sun 13	REST DAY	Review	-

Week 6 Schedule: July 14-20

Day	Morning (2 hrs)	Afternoon (1.5 hrs)	Evening (1 hr)
Mon 14	DAA: Heap Sort + Counting Sort (Abdul Bari)	Spring Boot: JWT Auth basics	College DSA + LC #200 Number of Islands
Tue 15	Trilogy session	DAA: Divide & Conquer intro	College DSA + LC #322 Coin Change
Wed 16	DAA: Greedy - Knapsack, Dijkstra's overview	Spring Boot: Deploy simple app on AWS free tier	College DSA + LC #62 Unique Paths
Thu 17	DAA: Dynamic Programming intro (Fibonacci, Climbing Stairs)	Revision: Java concepts	College DSA + LC #104 Max Depth Binary Tree
Fri 18	DAA: DP continued - 0/1 Knapsack	Revision: SQL + DBMS	College DSA + LC #15 3Sum

Sat 19	Full revision: DAA Unit I + II overview	Portfolio: Update GitHub README, organize repos	College DSA + LC #1143 LCS
Sun 20	BUFFER DAY	Prepare for semester	-

Complete LeetCode Tracker (34 Problems - 1 per day)

Phase 1: Trilogy Test Prep (June 16-22) - 7 problems

#	LC ID	Problem	Difficulty	Topic
1	#1	Two Sum	Easy	Arrays/HashMap
2	#206	Reverse Linked List	Easy	Linked List
3	#20	Valid Parentheses	Easy	Stack
4	#704	Binary Search	Easy	Searching
5	#125	Valid Palindrome	Easy	Strings
6	#53	Maximum Subarray	Medium	Arrays/Kadane's
7	#141	Linked List Cycle	Easy	Linked List

Phase 2: DBMS + DSA Grind (June 23 - July 6) - 12 problems

#	LC ID	Problem	Difficulty	Topic
8	#217	Contains Duplicate	Easy	Hashing
9	#242	Valid Anagram	Easy	Hashing/Strings
10	#121	Best Time to Buy/Sell Stock	Easy	Arrays
11	#88	Merge Sorted Array	Easy	Sorting
12	#26	Remove Duplicates Sorted Array	Easy	Arrays
13	#35	Search Insert Position	Easy	Binary Search

14	#509	Fibonacci Number	Easy	Recursion
15	#70	Climbing Stairs	Easy	DP/Recursion
16	#155	Min Stack	Medium	Stack Design
17	#75	Sort Colors	Medium	Sorting
18	#215	Kth Largest Element	Medium	Heap/Sorting
19	#198	House Robber	Medium	DP

Phase 3: Build + DAA (July 7-20) - 12 problems

#	LC ID	Problem	Difficulty	Topic
20	#55	Jump Game	Medium	Greedy
21	#3	Longest Substring No Repeat	Medium	Sliding Window
22	#33	Search Rotated Sorted Array	Medium	Binary Search
23	#49	Group Anagrams	Medium	Hashing
24	#78	Subsets	Medium	Backtracking
25	#46	Permutations	Medium	Backtracking
26	#200	Number of Islands	Medium	Graph/BFS
27	#322	Coin Change	Medium	DP
28	#62	Unique Paths	Medium	DP
29	#104	Max Depth Binary Tree	Easy	Trees
30	#15	3Sum	Medium	Two Pointer
31	#1143	Longest Common Subsequence	Medium	DP

NOTE: The college daily DSA challenge counts as your daily LeetCode. If the college problem is different from the one listed above, do the college problem first, then do the listed one if you have time. If not, skip the listed one - no stress.

Daily Routine (Realistic)

Good Day (4-5 hrs):

- Morning: 2 hrs - Primary task (Nimblize/DBMS/Spring Boot/DAA)
- Afternoon: 1.5 hrs - Secondary task
- Evening: 1 hr - College DSA challenge + LeetCode + push to GitHub

Busy Day (1-2 hrs):

- Evening only: College DSA challenge + 1 LeetCode problem
- Minimum viable progress - never zero

Tuesday (Trilogy Day):

- Trilogy session (time varies)
- Evening: College DSA + review what Trilogy covered

Sunday (Rest):

- No study. Light review at most. Recharge.
-

Rules (Updated)

1. **Java is your ONE language.** All DSA, all LeetCode, all projects - in Java. C++ only for DAA lab code (start in Phase 3).
 2. **College DSA challenge = your daily LeetCode habit.** Don't treat them separately.
 3. **Understand before you push.** If AI generates code for you, rewrite it yourself line by line before pushing to GitHub. If you can't explain every line, you didn't learn it.
 4. **Nimblize report this week is NON-NEGOTIABLE.** It's due Sunday. Do it first.
 5. **Trilogy test is your HIGHEST STAKES item.** Top 60 filtering. Prepare like it matters (it does).
 6. **Stop comparing paths.** Java + Backend + DB = your path. Everything else is a bonus layer.
-

By July 20, You Will Have

What	Proof
Java proficiency	30+ LeetCode problems solved in Java
DBMS knowledge	TransitPulse database (designed + implemented)
Backend understanding	Nimblize reports + Spring Boot basics

DSA foundation	Trilogy test + daily challenges + LeetCode profile
Cloud awareness	Nimblize cloud report + AWS free tier deployment
GitHub portfolio	Daily commits, DSA repo, TransitPulse project
Real-world experience	Nimblize startup reports, Trilogy training

That is MORE than enough foundation for placement prep by Jan 2027. You still have the entire 5th semester (Aug-Dec) to deepen everything.

Quick Links

Resource	Link
GFG Java	geeksforgeeks.org/java
GFG DBMS	geeksforgeeks.org/dbms
GFG SQL	geeksforgeeks.org/sql-tutorial
W3Schools SQL	w3schools.com/sql
Abdul Bari Algorithms	YouTube - Abdul Bari Algorithms playlist
Telusko Spring Boot	YouTube - Telusko Spring Boot playlist
LeetCode	leetcode.com
Draw.io (diagrams)	app.diagrams.net
AWS Free Tier	aws.amazon.com/free